

Extended exercises for Lecture "Theoretical Spectroscopy for systems implying periodic boundary conditions"**Exercise: Computing STM images****1. Computing an STM image for a molecule adsorbed on a slab**

- Start from a converged slab+molecule geometry. Cut out the top layer and the adsorbed molecule from your slab structure and use this one-layer structure for generating the STM images.

- Modify the input file by adding the following command lines in the &DFT section

```
&PRINT
&V_HARTREE_CUBE
STRIDE 12 12 12
&END
&MO_CUBES
NHOMO 1
NLUMO 1
STRIDE 12 12 12
&END
&END PRINT
```

- Run CP2K to generate the pd-v_hartree-1_0.cube file and further .cube files for the requested molecular orbitals.

- Based on these cube files, further Python scripts will compute STM images. To run the Python scripts, an exemplary bash script run_stm.sh is given in

```
/zfs/home/suphc356/TEACHING/TEACHING_1004D_WS202526/Inputs_Exercise6_STM
```

- The script is defining the following:

```
#!/bin/bash -l
conda activate /gxfs_home/cau/suphc356/.conda/envs/env_for_stm
FOLDER="/gxfs_work/cau/suphc356/stm/"
export FOLDER="/gxfs_work/cau/suphc356/stm/"
mkdir out

mpirun -n 128 python3 /gxfs_home/cau/suphc356/stm_tools/cp2k_stm_tools/stm_sts_from_wfn.py
-cp2k_input_file "$FOLDER"/cp2k.inp
-basis_set_file ../../BASIS_MOLOPT
-xyz_file "$FOLDER"/Acetophenon.xyz
-wfn_file "$FOLDER"/pd-RESTART.wfn
-hartree_file "$FOLDER"/pd-v_hartree-1_0.cube
-output_file "./out/stm.npz"
-orb_output_file "./out/orb.npz"

-eval_region "G" "G" "G" "G" "n-1.5_C" "p3.5"
-dx 0.15
-eval_cutoff 14.0
```

```
-extrap_extent 4.0
-p_tip_ratios 0.0

-n_homo 1
-n_lumo 1
-orb_heights 3.0 5.0
-orb_isovalues 1e-7
-orb_fwhms

-energy_range 2.5 3.5 0.05
-heights 4.5 4.4 4.2 4.3 4.1 4.6 4.7 4.8 4.9 5.0
-isovalues 1e-7 1e-6 1e-8 1e-5 1e-4 1e-3
-fwhms 0.5 0.4 0.3 0.2
cd out

/gxfs_home/cau/suphc356/stm_tools/cp2k_stm_tools/stm_sts_plotter.py -orb_npz orb.npz -stm_npz stm.npz
jobinfo
```

- Modify the bash script specifying the variable `FOLDER`, which should be set such that it is pointing to the directory where you performed the CP2K computation to compute the cube files.
- Submit the computation with sbatch:
`sbatch run_stm.sh`
- The generated STM images can be found in the so generated subfolder `stm`.